Fazleev Yan

MIPT
1st year Physics and Techology
School of Applied Mathematics and Computer Science
GPA (general): 9/10
GPA (informatics): 8.3/10

Contacts

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Skills

Programming:
C/C++, x86 assembly, Python, bash, Pascal

Tools:
Git, Make/Cmake, radare2, LaTeX, Doxygen, Graphviz, jq, SQL, valgrind, docker, SIMD

Spheres of Interests

- Compilers & optimizations
- Low-level programming
- Operating systems

Personality

- Team-work
- Sociable
- Creative
- Adaptable
- Fast-learning
- Hard-working

Projects

Virtual Processor (SPU): October 2024

Tools: C, Make

Emulation of processor and assembly instruction interaction. There are two stages of processing: reading instructions and converting it to bytecode and then processing this bytecode. Video memory is also implemented.

Programming Language: December 2024 - now

Tools: C, Graphviz, Make

My programming language. The code goes through the tokenization stage, parsing (using recursive descent), is translated into a syntax tree with a table of names, and then processed by the compiler using SPU.

MyPrintf: March 2025

Tools: C, assembly x86, radare2, Make

My implementation of the printf function. Using x86 assembly and C. The MyPrintf function supports 7 specifiers: %(%|b|o|x|d|c|s).

Mandelbrot Set: March 2025

Tools: C/C++, SIMD, Python, Make

Applying SIMD instructions to optimize the rendering of the Mandelbrot set (using avx2). In result, it turned out to speed up 7 times.

Hash Table: April 2025 - now

Tools: C, SIMD, x86 assembly, valgrind, Make

Hash table that uses a linked list to handle collisions. Various hash functions have been compared. Study of profiles and further optimizations of "hot" functions.

Other

Languages: Russian, English

Hobbies: volleyball, books, music, intellectual games